ALAKH SIR ke Fareey

ACIDS, BASES AND SALTS

ACID :- A substance that gives $H^+(aq)$ as only +ve ion in aqueous solution.

Example :- HCl, H_2SO_4, HNO_3
CH_3COOH (acetic acid)

(i) Acids are sour (khatte) in taste
(ii) Acids turns blue litmus red.

BASE :- A substance which increase Hydroxyl ions $OH^-(aq)$ in aqueous solution.

Example- $NaOH$, KOH, NH_4OH

(i) Bitter in taste.
(ii) Soapy to touch.
(iii) Base turns red litmus blue.

INDICATORS :- Tells wheater a substance is acid or base.

Indicators	Base	Acids
Litmus	Blue	Red
Turmeric	Red	yellow
Phenolphthalein (Colorless)	Pink	colourless
Methyl Orange Range	yellow	Red

Trick -> LaBouR TRY PoPCorn with MaYoR

(i) Litmus solution is originally purple in colour where no acid or base is added.
(ii) Pure water has no effect on colour of litmus.

OLFACTORY INDICATORS :- substance whose smell odour changes in acidic or basic medium.

Example :- Onion, vanilla essence, clove oil

Acid Retains smell

Base loses smell

ACID AND BASE IN WATER

Acids produce H^+ ions in aqueous Solution :-

$H_2SO_4 \xrightarrow{H_2O} 2H^+(aq) + SO_4^{2-}$

$HCl \xrightarrow{H_2O} H^+(aq) + Cl^-$

$H^+ + H_2O \longrightarrow H_3O^+$ (Hydronium ion)

$H^+(aq) = H_3O^+$ gives common properties to all acids.

Preparation of HCl gas :-

$2NaCl(aq) + H_2SO_4(conc.)$
$\downarrow$
$Na_2SO_4(aq) + 2HCl(g)$

$HCl(g) \xrightarrow[\text{litmus}]{\text{dry blue}}$ No change

$HCl(g) \xrightarrow[\text{litmus}]{\text{moist blue}}$ Red colour

Alkalis - Some bases are water soluble, these are called Alkalies.

$NaOH(s) \xrightarrow{H_2O} Na^+(aq) + OH^-(aq)$

$KOH(s) \xrightarrow{H_2O} K^+(aq) + OH^-(aq)$

Example :- $NaOH$, KOH, $NH_4(OH)$

- All Bases are not water soluble.
- Bases but not alkali - $Zn(OH)_2$, $Fe(OH)_2$, $Cu(OH)_2$

Acid + water is a highly Exothermic.

Always add acid slowly to water with constant stirring.

if water is added to a concentrated acid, the heat generated may cause the mixture to splash out and cause burns. The glass container also break due to heat given out.

Trick

AaThuW
Acid to water

Acid and Alkali - Electric current ?

- Electric current through the solution is carried by ions.
- Solution of Acids :- HCl, H_2SO_4, HNO_3 and CH_3COOH generates ions and hence they conduct electricity.
- Alkalis also generate ions - $NaOH$, KOH, $Mg(OH)_2$, NH_4OH and hence conduct electricity.
- Glucose, Alcohol do not generate ions and hence do not conduct electricity.

REACTION with METAL

Metal + Acid(dil) $\longrightarrow$ Salt + $H_2\uparrow$

	Most Reactive ↑	
K (Potassium)		Katrina
Na (Sodium)		Ne
Ca (Calcium)		Car
Mg (Magnesium		Mangi
Al (Aluminium)		Alto
Zn (Zinc)		Zen
Fe (iron)		Ferrari
Pb (lead)		Firbi
H (Hydrogen)		Hath
Cu (Copper)		kyu
Hg (Mercury)		Mili
Ag (Silver)		Silver
Au (Gold)		Audi
	less Reactive	

Preparation of Hydrogen gas

$H_2SO_4 + Zn \longrightarrow ZnSO_4 + H_2\uparrow$ (White Vitriol)

- $ZnSO_4$ is white coloured salt called White vitriol.
- H_2 gas burns with pop sound and Extinguishes a candle.
- Combustible but not supporter of Combustion.

✡ Metal + Base $\longrightarrow$ Salt + H_2

$2NaOH(aq) + Zn(s) \longrightarrow Na_2ZnO_2(s) + H_2\uparrow$ (sodium zincate)

- Does not happen with all metals.

NEUTRALISATION :-

Acid + Base $\longrightarrow$ salt + water

- $HCl(aq) + NaOH(aq) \longrightarrow NaCl + H_2O$
- $H_2SO_4 + Ca(OH)_2 \longrightarrow CaSO_4 + H_2O$

Metal oxide + Acid $\longrightarrow$ salt + water
(ZnO/CuO/MgO/CaO)

$CuO(s) + 2HCl(dil) \longrightarrow CuCl_2(aq) + H_2O$
(Black solid) (Blue-green)

- Generally Metal oxide are basic in nature because they react with Acid to give salt and water.

Non Metal oxide + Base ⟶ salt + water
($SO_2/SO_3/CO_2/NO_2$)

$CO_2 + Ca(OH)_2\,aq \longrightarrow CaCO_3 + H_2O$

- Generally Non-metal oxide are acidic in nature because they react with base to give salt and water.

Acid + Metal carbonate (CO_3^{2-})
OR
Metal Hydrogen carbonate HCO_3^-
(Bicarbonate)
↓
salt + CO_2 + H_2O↑

CO_2 turns lime water Milky

- $CO_2(g) + Ca(OH)_2 \longrightarrow H_2O + CaCO_3(s)\downarrow$
(lime water) (White ppt) (Calcium carbonate)

Test :-
on passing excess of CO_2, milkiness disappears.

- $CaCO_3 + H_2O + CO_2 \longrightarrow Ca(HCO_3)_2(aq)$

$Na_2CO_3 + 2HCl \longrightarrow 2NaCl + H_2O + CO_2$

Strength of Acid and Base

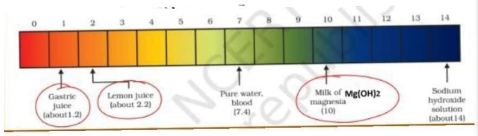
pH : P (Potenz → Power)
H → Hydrogen

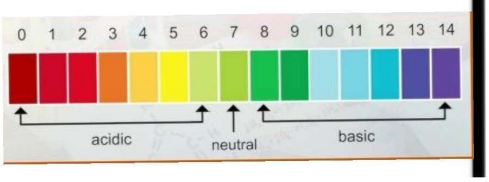
- Measure $H^+(aq)$ ions concentration in a solution.
- pH ranges from 0 (very acidic) to 14 (Very basic)
- pH↓ → $H^+(aq)$↑ → Acidic↑

pH < 7 Acidic
pH = 7 Neutral
pH > 7 Base

Universal indicator :-

Substance which detect nature of chemical as acid or base and also measure strength of it. eg- pH paper





Strong Acid :- H_2SO_4, HCl, HNO_3
- Gives more concentration of $H^+(aq)$ ions.

Weak Acid :- CH_3COOH (Acetic acid), citric acid, lactic acid, H_2CO_3 (carbonic acid).
(generally organic acids - natural sources)
- Gives less concentration of $H^+(aq)$ ions.

Strong Base :- NaOH, KOH, $Ca(OH)_2$
- Gives more concentration of OH^- ions.

Weak Base :- NH_4OH, $Zn(OH)_2$, $Cu(OH)_2$, $Fe(OH)_2$.
- Gives less concentration of OH^- ions.

pH of salts ?

Acid + Base ⟶ salt + Water

✓ Strong Acid (HCl) + Strong Base (NaOH) ⟶ Neutral salt (NaCl) + water (PH=7)

✓ Weak Acid (H_2CO_3) + Strong Base (NaOH) ⟶ Basic salt (Na_2CO_3) + Water (PH>7)

✓ Strong acid (HCl) + Weak Base (NH_4OH) ⟶ Acidic salt (NH_4Cl) + Water (PH<7)

Naturally Occuring Acids

Natural Source	Acid	Trick
Vinegar ⟶	Acetic Acid ⟶	sirka pikar Acting karne lage.
Orange ⟶	citric Acid ⟶	orange cheela andar city nikli
Tamarind ⟶	Tartaric Acid ⟶	Imlee ko tar se bandha.
Tomato ⟶	Oxalic Acid ⟶	Tamatar OX ne khaya
sour milk (curd) ⟶	lactic Acid ⟶	Dahi ko lake me feka.
lemon ⟶	citric Acid ⟶	lemon cheela andar city nikli
Ant sting ⟶	Methanoic acid ⟶	Ant ne khaya Methi ka paratha.
Nettle sting ⟶	Methanoic acid ⟶	Net me fasa Methi ka paratha

Common salt (NaCl) [Rock salt] :-

(1) sodium Hydroxide / Caustic soda (NaOH)

- Chlor - Alkali process

$2NaCl(aq) + 2H_2O(l) \xrightarrow{\text{electric current}} 2NaOH(aq) + Cl_2(g) + H_2(g)$
Brine (near cathode) (at anode) (at cathode)

Chor khali
PAr Cl → Positive Anode chlorine
CHiNa → cathode Hydrogen NaOH

- found in deposits in rock bed.
- Raw material for various chemical.

(2) Bleaching powder ($CaOCl_2$)

$Ca(OH)_2 + Cl_2 \longrightarrow CaOCl_2 + H_2O$
(Slaked lime) (Bleaching Powder)

Uses :-
(i) Bleaching cotton and linen, wood pulp in paper factories, Bleaching washed clothes.
(ii) Oxidising agent.
(iii) Make drinking water free form germs.

(3) Baking Soda ($NaHCO_3$)

$NaCl + H_2O + CO_2 + NH_3$
↓
$NaHCO_3 + NH_4Cl$
(sodium Hydrogen carbonate baking soda)

Uses :-
(i) for making tasty crispy pakora.
(ii) for faster cooking
(iii) Neutralise Acidity Antacid.
(iv) Soda-Acid fire extinguisher
(v) Delay curdling of milk.

Baking powder :-
Baking powder = Baking soda + Tartic Acid
$NaHCO_3 + H^+ \longrightarrow CO_2 + H_2O$ + salt
Uses :-
- cause bread or cake to rise making them soft or spongy.

(4) Washing Soda ($Na_2CO_3 \cdot 10H_2O$)

$2NaHCO_3 \xrightarrow{\text{Heat}} Na_2CO_3 + H_2O + CO_2$

$Na_2CO_3 + 10H_2O \longrightarrow Na_2CO_3 \cdot 10H_2O$

Uses :-
(i) Glass, soap and paper industry.
(ii) Removing permanent hardness.

WATER OF CRYSTALLIZATION :-

- Fixed number of water molecules present in one formula unit of salt.
- $CuSO_4 \cdot 5H_2O$ (copper sulphate Crystals)
- $FeSO_4 \cdot 7H_2O$ (Ferrous sulphate Crystals)
- $CaSO_4 \cdot 2H_2O$ (Gypsum)
- $Na_2CO_3 \cdot 10H_2O$ (washing soda) Sodium carbonate decahydrate)

✡ $CuSO_4 \cdot 5H_2O \xrightarrow{\text{Heat}} CuSO_4 + 5H_2O$
(Blue) (white)

✡ $FeSO_4 \cdot 7H_2O \xrightarrow{\text{Heat}} FeSO_4 \xrightarrow{\text{Heat}} Fe_2O_3$
(Green) (White) (Brown)

Plaster of Paris :- (POP)

$CaSO_4 \cdot \frac{1}{2}H_2O$ —— White powder

▲ $CaSO_4 \cdot 2H_2O \xrightarrow{373k} \frac{3}{2}H_2O + CaSO_4 \cdot \frac{1}{2}H_2O$
(Gypsum) (POP)

✡ Plaster of paris is used for making toys, making surface smooth and materials of decoration.

CURRENT YEAR QUESTIONS

QUESTION-1) Riya was helping her brother clean up their old toolbox when she noticed a few pieces of zinc metal. Curious about its properties, she remembered a science lesson where zinc reacts with certain solutions to produce gas. she decided to recreate the reaction at home safely.

Riya placed a few pieces of the zinc metal in a small container and poured a sodium hydroxide solution over it. After gently warming the mixture, she noticed bubbles forming, which she collected using a soap solution. she then tested the gas to understand its properties.

later, she wondered which gas would form if zinc reacted with a strong acid instead of sodium hydroxide.

(a) Write the equation of the chemical reaction involved and the test to detect the gas.

(b) Name the gas that will evolve when the same metal reacts with a dilute solution of strong acid. (CBSE 2015, 2018, 2024) [CBQ]

QUESTION-2) A student took a small amount of copper oxide in a conical flask and added dilute hydrochloric acid to it with constant stirring. He observed a change in the colour of the solution.

(a) Write the name of the compound formed and its colour.

(b) Write a balanced chemical equation for the reaction involved. (CBSE 2016, 2021, 2023)

QUESTION-3 An aqueous solution 'A' turns phenolphthalein solution pink. when another aqueous solution 'B' is added to the pink solution, the pink colour disappears. Now When a few drops of solution 'A' are added to this reaction, the mixture appears pink again. The respective changes in the nature of the solution are from.

(a) acidic ⟶ basic ⟶ basic (CBSE 2021, 2022, 2023, 2024) [CBQ]

(b) basic ⟶ acidic ⟶ acidic

(c) acidic ⟶ basic ⟶ acidic

(d) basic ⟶ acidic ⟶ basic

QUESTION-4) Anika's grandmother was preparing pickles at home when she accidentally spilled some pickle juice on her hand. she noticed a sharp change in the smell of her hands. Intrigued, Anika rembembered a science project about natural indicators like onion juice. she decided to test the pickle juice, calling it liquid 'X' with different materials.

she discovered that –

liquid 'X' slightly changed the odour of a cloth strip dipped in onion juice.

liquid 'X' also turned blue litmus paper red, indicating it had a specific chemical property.

To further understand the behaviour of liquid 'X', Anika tested it with.

(a) zinc granules (b) solid sodium carbonate

(CBSE 2019, 2020, 2023) [CBQ]

QUESTION-5) The pH of a salt used to make tasty and crispy pakoras is 9. identify the salt and write a chemical equation for this formation. List its two uses (CBSE 2018, 2019, 2020, 2023, 2024)

QUESTION-6) (i) Define water of crystallisation.

(ii) Write the chemical formula of bleaching powder.

(iii) Write the chemical name and formula of a compound having water of crystallisation in its molecule and appears blue.

Write a balanced chemical equation of the reaction involved in its preparation. List its three uses. (CBSE 2015, 2020, 2023)

QUESTION-7) How is washing soda prepared from sodium carbonate? Give its chemical equation. state the type of this salt. Name the type of hardness of water which can be removed by it. (CBSE 2017, 2020, 2023)

QUESTION-8) During a visit to a glass factory. Meera noticed workers handling a white powdery substance labeled 'X' that was being used in the production process. curious, she asked about its uses and learned that this compound is not only important in glassmaking but also plays a role in the soap industry. she also discovered that it is prepared from a saltwater solution called brine.

later, Meera decided to research more about this compound. while studying, she came across an interesting reaction of this substance with water containing salts of calcium or magnesium.

Based on the above context, answer the following questions.

(a) What is the chemical name, common name, and chemical formula of compound 'X'?

(b) Write the chemical equation for the preparation of compound 'X' from brine.

(c) What is the name given to this to this reaction? Are there any gases evolved durning this reaction. What is the location of gases.

(CBSE 2015, 2018, 2020) [CBQ]

QUESTION-9) Ravi was helping his father renovate their house when he noticed a worker mixing a white powder with water to repair wall cracks. Curious Ravi asked what the powder was, and the worker explained that it was plaster of paris (POP), a material widely used in construction and decoration. Ravi then remembered learning that POP is made from a mineral called Gypsum.

Ravi wanted to understand more about how POP is obtained and its applications in daily life.

Answer the following Questions:

(a) How is plaster of Paris obtained from gypsum? Write the chemical equation for the process.

(b) List any two uses of plaster of paris. (CBSE 2016, 2017, 2020, 2024) [CBQ]

QUESTION-10) When a few drops of universal indicator are added to three colourless solutions, X, Y, and Z. they produce green, red and blue colours, respectively.

(a) Arrange the solutions X, Y, Z in increasing order of PH.

(b) identify which of the three solutions — X, Y or Z — will change the colour of phenolphthalein, and explain the reason for this behavior. (CBSE 2019, 2019C)

QUESTIONS-11) What happens when sulphur is burned, and the fumes are collected? if water is added to the test tube containing the fumes and shaken well, how will the resulting solution behave with blue and red litmus paper? Is the product acidic or basic? What would happen if hydrogen chloride gas (HCl) is treated with dry and moist blue litmus. which ion is responsible for the colour change in the case? (NCERT)

✡✡ Asked in different format in exam.